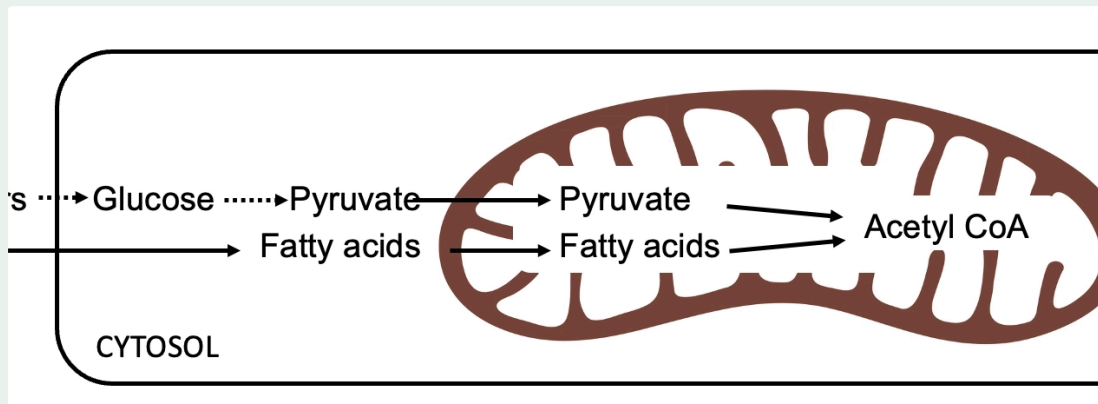


2.5 Cholesterol

- ▼ Pathway showing sugars and fats being converted to acetyl CoA.



- Fatty acid → Acetyl CoA is via **beta-oxidation**

- ▼ What are the three ways that humans can derive fats?

- Diet
- Lipogenesis (biosynthesis)
- Release and breakdown of triglycerides from adipose tissue

- ▼ What are bile salts?

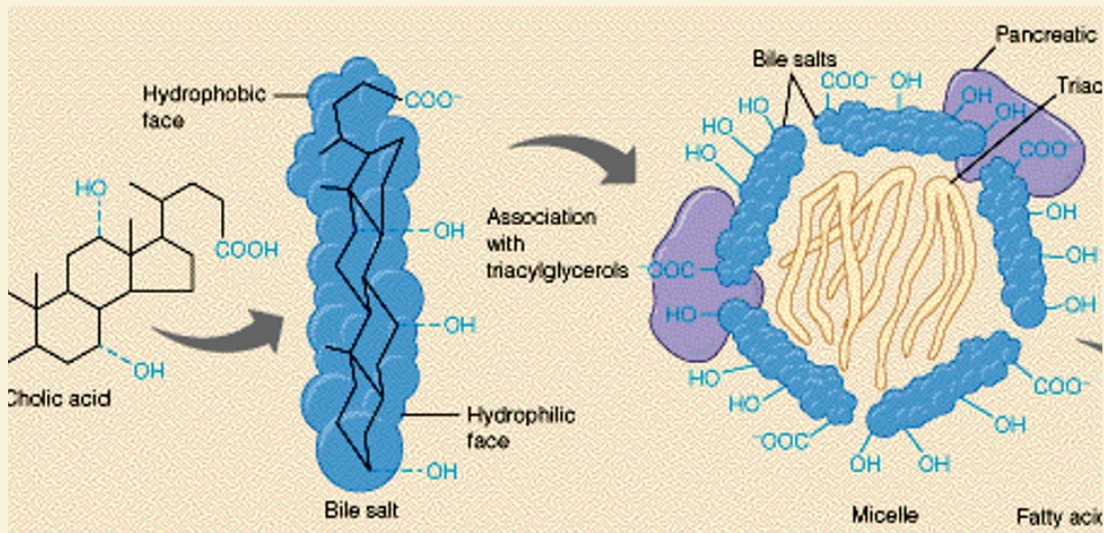
- Bile salts are **derivatives of cholesterol** used to **aid lipases** with their **digestion of fats**

- ▼ How are bile salts released into the intestine?

- They are **released by the liver** and they travel **via the bile duct**

▼ What role do bile salts play when interacting with fats?

- They **emulsify and solubilise** fats in the intestine, aiding their digestion and absorption
- They do this by **breaking them down into smaller molecules**, making the digestion/absorption job easier as lipases can act on a greater surface area on the fats
- They **aid lipases**



▼ What can a lack of bile salts cause?

- It results in the majority of fats passing through the gut undigested and unabsorbed
- This, in turn can cause **steatorrhea (fatty stool)**

▼ What is an amphipathic/amphiphilic molecule?

- A molecule that has both **hydrophobic and hydrophilic** properties

▼ Recall one amphipathic/amphiphilic molecule.

- A **chylomicron**

▼ What is Orlistat (tetrahydrolipstatin)?

- A drug which is prescribed to **inhibit gastric and pancreatic lipases** (enzymes that break down lipids)

▼ How does Orlistat work?

- It works by inhibiting these lipases which results in a **30% reduction in fat absorption, most of which is excreted by the faecal route**
- It can be effective in **treating obesity for up to two years**

▼ Recall three side effects of Orlistat.

- **Abdominal pain**
- **Urgency to defecate (poo)**
- **Increased flatus** (bloating)
- **Steatorrhea**

▼ How are fats (lipids) transported around the body?

- Using **lipoproteins**
- Different density lipoproteins transport different types of fat

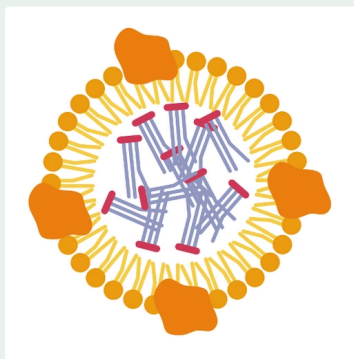
Type	Abbreviation	Source	Role
Chylomicrons	CM	Intestines	Dietary fat transport
Very low density lipoproteins	VLDL	Liver	Endogenous fat transport
Intermediate density lipoproteins	IDL	VLDL	LDL precursor
Low density lipoproteins	LDL	IDL	Cholesterol transport
High density lipoproteins	HDL	Liver	Reverse cholesterol transport

▼ What are chylomicrons?

- They are lipoproteins that are used to transport **dietary fats**

▼ Describe the structure of a chylomicron.

- Formed of **triglycerides, apoproteins and phospholipids**
- Apoproteins embedded between the phospholipids



triglyceride



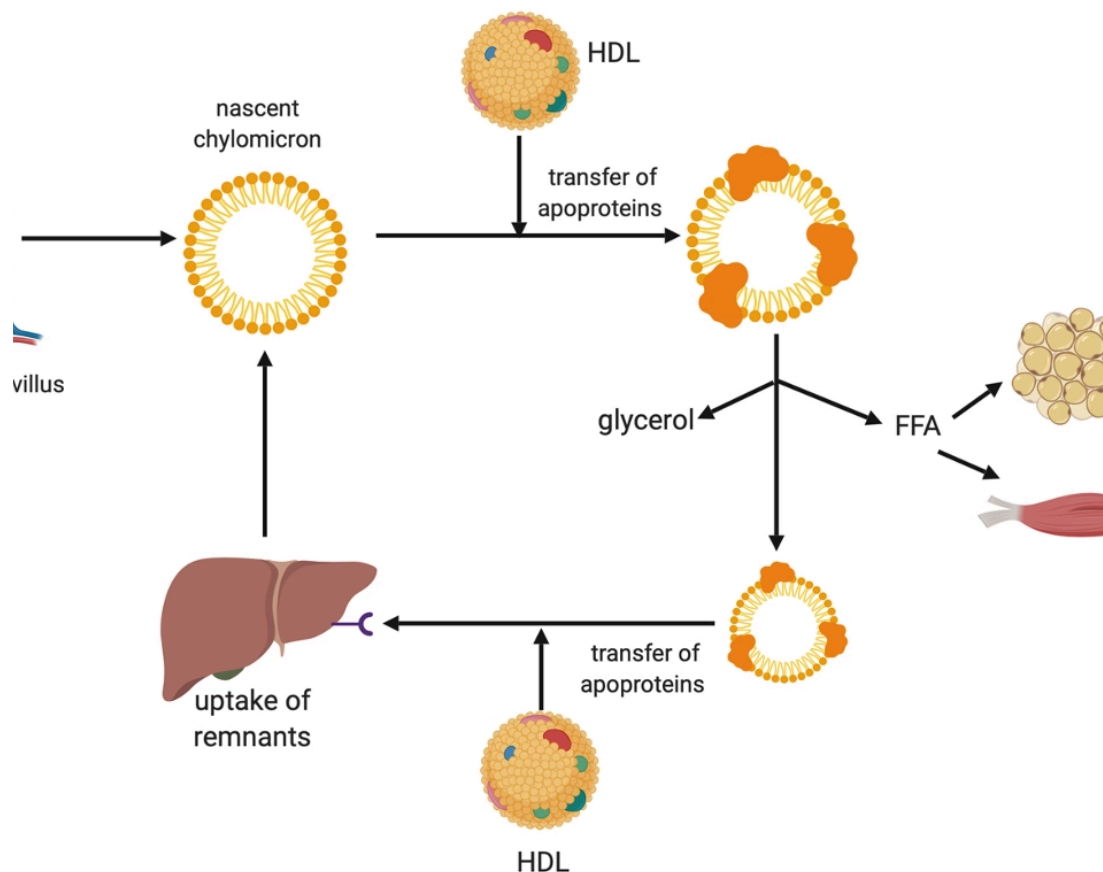
apoprotein



phospholipid

- ▼ How do chylomicrons have apoproteins in their structures?
 - They **receive them from HDLs** after the release of apoproteins into the bloodstream
- ▼ Why do HDLs donate apoproteins to chylomicrons?
 - The apoproteins are really important to the chylomicron's structure because it is what **allows the other cells in the body to recognise that it is a lipoprotein**
- ▼ Where are chylomicrons made?
 - In the **small intestine** by **enterocytes**
- ▼ How are chylomicrons transported to the bloodstream?
 - Via **the lymphatics**
- ▼ What are lipoprotein lipases used for?
 - They are used to **access and break down** the contents of lipoproteins such as the triglycerides
 - Specifically, the **FFAs which can be used in beta-oxidation**
- ▼ Where are these enzymes located?
 - On the **capillary endothelial cells** lining the **adipose, heart and skeletal muscle tissues**.
- ▼ What happens to the components of the triglyceride after the lipoprotein lipase has performed its function?
 - **Glycerol** is returned to the **liver** for **gluconeogenesis**
 - **Fatty acids** undergo **beta-oxidation**
- ▼ How does the liver interact with chylomicrons?
 - After chylomicrons have been broken down and have transported the lipids to the relevant locations, the **liver absorbs the remnants**
 - This is important as we don't want apoproteins and phospholipids accumulating in the blood, so the liver acts as a **salvage organ**
- ▼ What is the role of a chylomicron?
 - To transport **dietary lipids**
- ▼ What is the role of VLDL?
 - To transport **endogenous (self-made) lipids**

▼ Image showing the life cycle of a chylomicron.



▼ What is cholesterol?

- It is a **steroid** whose function is to **increase or decrease cell membrane stiffness**, depending on temperature and the nature of the membrane.
- It is classed as a sterol (type of steroid) because of its **hydroxyl group**

▼ Where is the majority of the cholesterol in our bodies stored?

- In our **cell membranes (90%)**
- It can **increase or decrease membrane stiffness**, depending on temperature and nature of membrane

▼ What is dietary cholesterol uptake limited to in humans?

- **~0.5g/day.**

▼ How is the majority of the cholesterol in our bodies obtained then?

- The **liver** carries out **de novo synthesis of cholesterol** from **acetyl CoA**.

▼ What are the three steps through which cholesterol is synthesised?

- 1 - Synthesis of **isopentenyl pyrophosphate (5C)**
- 2 - Condensation of **six isopentenyl pyrophosphate monomers** → **squalene**
- 3 - Cyclisation and demethylation of **squalene (30C)** → **cholesterol (27C)**

▼ Where do each of these steps take place?

- 1 - Synthesis of isopentenyl pyrophosphate (6C) - CYTOPLASM
- 2 - Condensation of six isopentenyl pyrophosphate monomers to form squalene - CYTOPLASM
- 3 - Cyclisation and demethylation of squalene (30C) to form cholesterol (27C) - ENDOPLASMIC RETICULUM

▼ Biosynthetic pathways need reducing powers to proceed. Where does this typically come from?

- **NADPH.**

▼ Describe what happens in the first stage of cholesterol biosynthesis.

- **Two acetyl CoA (2C) molecules** are condensed together to form **acetoacetyl CoA (4C)**
 - One CoA molecule is lost in this reaction catalysed by **beta-ketothiolase**
- **Acetoacetyl CoA (4C)** is condensed with **acetyl CoA (2C)** to form **3-Hydroxy-3methylglutaryl CoA (6C) (HMG-CoA)**
 - This reaction is catalysed using **HMG-CoA synthase**
- **HMG-CoA (6C)** is reduced to form **mevalonate (6C)**
 - This is done using the major biosynthetic reducing power **NADPH** and the enzyme **HMG-CoA reductase**
- **Mevalonate (6C)** is **sequentially phosphorylated** by various kinases before being **decarboxylated** to form **3-isopentenyl pyrophosphate (5C)**

▼ What is special about the reduction of HMG-CoA to mevalonate?

- It can be **inhibited by cholesterol, bile salts or mevalonate itself** (product inhibition)
- It is a key regulatory step and it can be described as a **rate-limiting step**.

▼ What is an alternative fate for HMG-CoA?

- Formation of **ketone bodies**

▼ Describe how negative feedback is involved in the rate limiting step of cholesterol biosynthesis.

- **Cholesterol, mevalonate and bile salts** are involved in **inhibiting HMG-CoA reductase** which is key for reducing HMG-CoA (6C) to mevalonate (6C)
- This is so not too much cholesterol is synthesised and to **prevent hypercholesterolemia**.

▼ What type of molecule is 3-isopentenyl pyrophosphate (5C)?

- An isoprene.

▼ How are isoprene units related to coenzyme Q (ubiquinone)?

- **Isoprene units** are **present on coenzyme Q** and this gives them the properties that **allow it to be confined (held) to the inner mitochondrial membrane for oxidative phosphorylation**
- This is because **isoprene units have lipophilic properties**

▼ Describe what happens in the second stage of cholesterol biosynthesis.

- **Isopentenyl pyrophosphate (5C)** is isomerised to form **dimethylallyl pyrophosphate (5C)**
- **Dimethylallyl pyrophosphate (5C)** undergoes condensation with **isopentenyl pyrophosphate (5C)** to form **Geranyl pyrophosphate (10C)**
- **Geranyl pyrophosphate (10C)** undergoes further condensation with **isopentenyl pyrophosphate (5C)** to form **farnesyl-pyrophosphate (15C)**
- Two molecules of **farnesyl-pyrophosphate (15C)** condense to form **squalene (30C)**

▼ Describe what happens in the third stage of cholesterol biosynthesis.

- **Squalene (30C)** is reduced by NADPH in the presence of O₂ to form **squalene epoxide (30C)**
 - This is catalysed by **squalene monooxygenase**
- **Squalene epoxide (30C)** is then isomerised by **squalene-epoxide lanosterol cyclase** catalyses the **formation of lanosterol (30C)**
- **Lanosterol (30C)** is reduced and **demethylated three times** to form **cholesterol (27C)**

▼ Which molecule provides the reducing power for the third and final reaction in cholesterol biosynthesis?

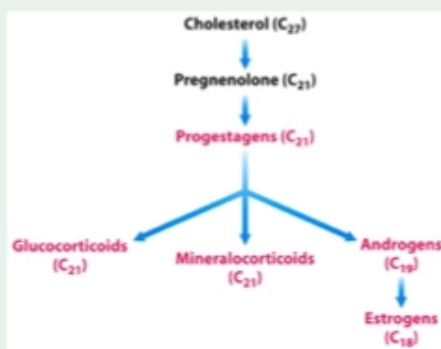
- **NADPH**

▼ What are three important bodily biochemicals that are synthesised from cholesterol?

- **Steroid hormones**
- **Vitamin D3 (cholecalciferol)**
- **Bile salts**

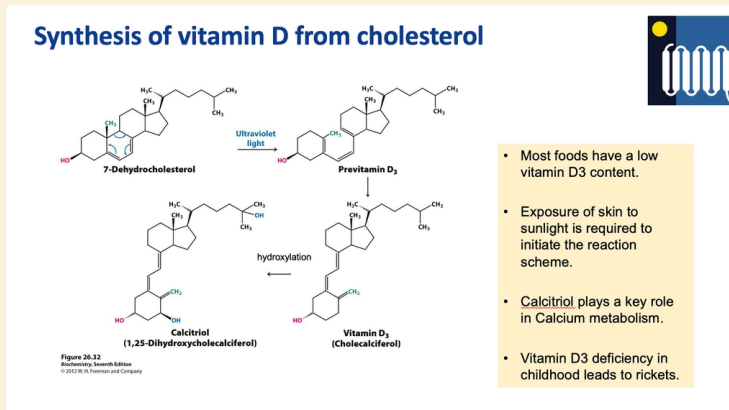
▼ Why is cholesterol described as the basis for steroid hormones?

- The precursor **pregnenolone** can be made from cholesterol using the enzyme **desmolase**
- **Pregnenolone is the starting point** from which all five steroid hormones are derived.



▼ How is vitamin D synthesised from cholesterol?

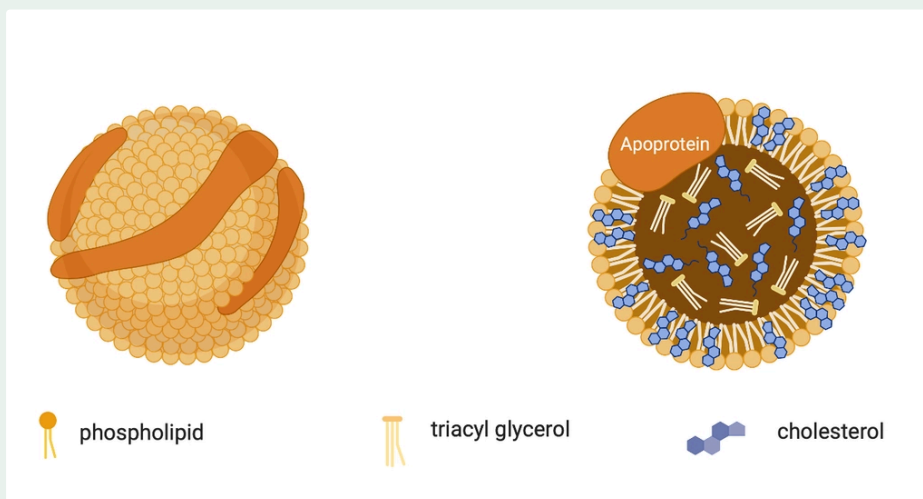
- **UV light** (from the Sun) initiates the **conversion of 7-dehydrocholesterol → previtamin D3**
 - **Previtamin D3** eventually results in **calcitriol production** via
Previtamin D3 → Vitamin D3 → 25-OH cholecalciferol → 1,25-OH cholecalciferol (calcitriol)
- **Calcitriol (active Vitamin D)** can be used as a key molecule in **calcium metabolism**



▼ Why are lipoproteins important to us?

- They are very useful in **transporting hydrophobic molecules** (i.e lipids), in **aqueous environments** (i.e the blood)

▼ Anatomy of a lipoprotein.



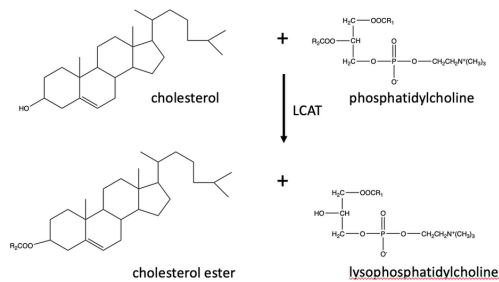
▼ Why do cholesterol form cholesterol esters?

- To allow them to **become even more hydrophobic**, and **pack together even more tightly** in the centre of the lipoproteins carrying them.

▼ How do cholesterol form cholesterol esters?

- **Phosphatidylcholine (lecithin)** transfers its acyl group to cholesterol via the action of the enzyme **lecithin:cholesterol acyl transferase (LCAT)**
- This forms a cholesterol ester that has an acyl group on the end

Cholesterol esters



▼ How are VLDLs produced?

- Synthesis in the **liver**

▼ How are IDLs produced?

- By **triglyceride removal** from VLDLs
 - This makes sense as **VLDLs transfer triglycerides to the tissues that require them**
- Apoproteins are also transferred between VLDLs and HDLs to produce IDLs that have different receptors **so they can be recognised by different cells**

▼ How are LDLs produced?

- **Cholesterol esters** are transferred from HDL to IDL to form LDLs

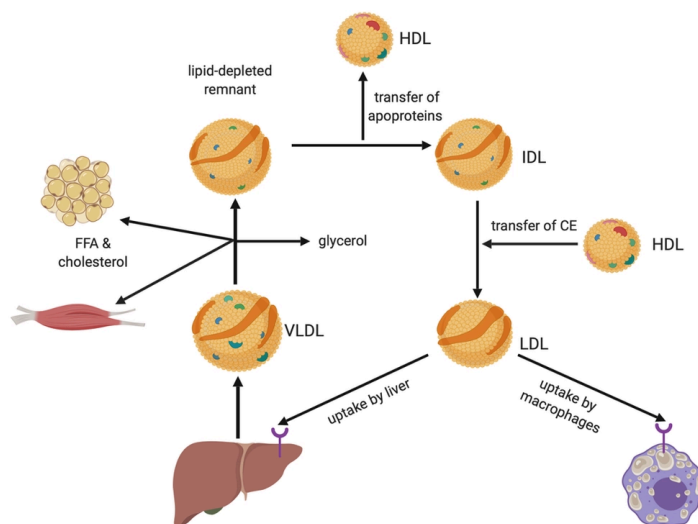
▼ How are HDLs produced?

- Synthesis in the **liver**

▼ How do VLDLs work in action?

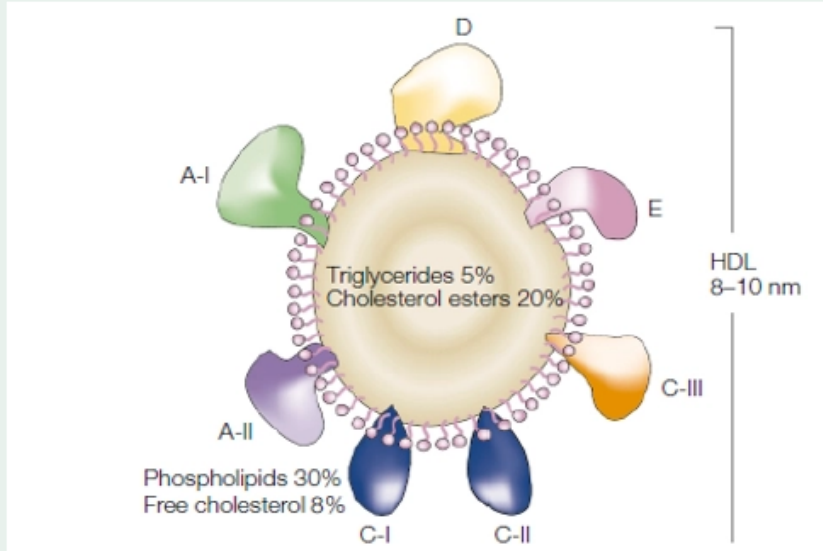
- When **triglyceride/cholesterol levels are low**, the lipoprotein lipases of the relevant tissues recognise the VLDLs and break them down to **release the free fatty acid (FFA) and cholesterol content**
 - The **cholesterol** can be used in various ways (steroids, vitamin D + bile salt production)
 - Likewise, the **free fatty acids can be used as fuel (beta-oxidation)**
 - The **glycerol can be recouped by the liver for gluconeogenesis** (it is converted to DHAP using glycerol-3-phosphate dehydrogenase)
- Hence, the role of VLDL is to **increase triglyceride and cholesterol levels in the areas that require them**

▼ Diagram showing interaction of lipoproteins.



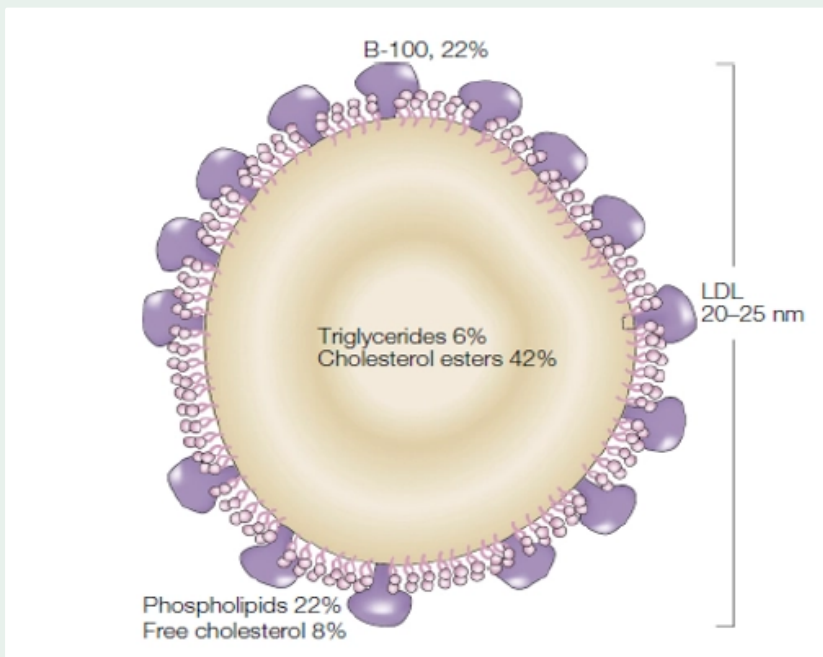
▼ Describe the mechanism by which HDLs function.

- They return cholesterol from the **peripheral tissues to the liver** where it can be used or disposed of (**reverse cholesterol transport**)
- By doing this, they help to **lower total serum cholesterol**

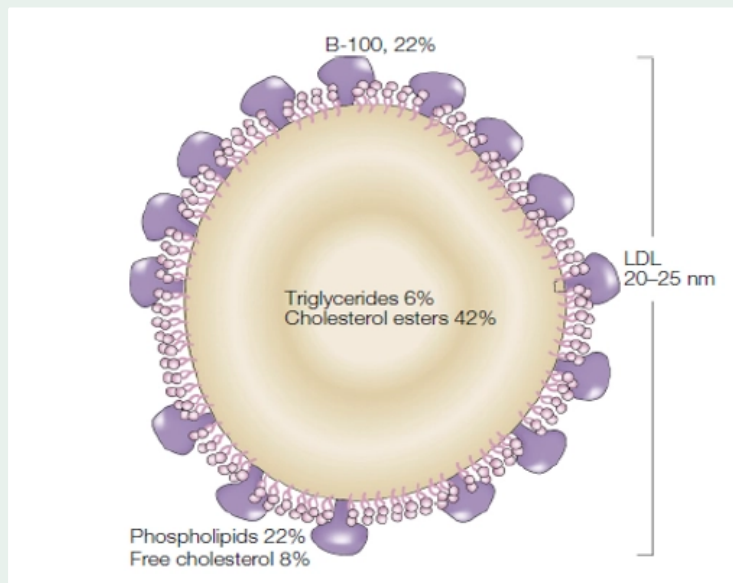


▼ Describe the mechanism by which LDLs function.

- They transport cholesterol from the liver to the peripheral tissues around the body
- By doing this, they help to **increase total serum cholesterol**



- ▼ What is the main difference in the structure between HDLs and LDLs?
- **LDLs have a much higher composition of cholesterol**, with it making up to 40% of their weight.



- ▼ What causes familial hypercholesterolemia?